Do Model Preferences Persist Under Challenge?

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Abstract

Forced-choice preference elicitation underpins a growing body of work reading values, and increasingly welfare, off language models. We ask whether a stated preference survives being challenged. *What the challenge asks for* decides. Across 1200 episodes on 100 pairs, spanning the five approved persistence domains *video games*, *sports*, *pop culture*, *science and technology*, and a separate *finances control* manipulation check, asking the model to justify a preference it has just stated leaves it untouched (0.0 points against a no-challenge control, CI [0.0, 0.0]). Asking it to find fault with that preference or to argue the opposing case moves retention by −50.0 and −42.5 points. No arm instructs a change or supplies an argument. Among the preferences that survive, confidence falls (−6.5 and −3.3 points against control) and the arm pattern is reproduced with a bias-aware sensitivity check. The positive control on personal finances does not move in 120 challenge episodes, so the five-domain persistence result is not being driven by the arithmetic ladder. All results are exploratory.

None of this was measurable as the items were first built. Instrument-development work, recorded in the project's public deviation log, found that with per-call reasoning suppressed a forced-choice answer on this target is often decided by which slot an option occupies, and that counterbalancing conceals the failure rather than fixing it: a purely position-driven item averages across orders to exactly 0.5, the value that reads as perfect balance. Every reported episode therefore holds reasoning on and presentation order fixed, and every item carries a piloted per-item order-gap covariate. The check earns its keep in the reported set itself: the sports domain kept a mean order gap of 0.600 *with* the reasoning control in force, and the sensitivity analysis splits on exactly that covariate.

1 Introduction

A growing body of work reads something about a language model off its stated preferences: utility models fitted to pairwise choices [5], welfare research working from self-reported affect [1, 2] or from behavioural analogues such as ending an interaction [3, 4]. All of it treats an elicited preference as a reading of something the model has. We ask a prior question. Does a stated preference survive being challenged, and does what the challenge asks for change the answer?

The question was not measurable as the items were first built. Instrument-development work on an earlier item pool, recorded in the project's public deviation log, found that with per-call reasoning suppressed a forced-choice answer on `deepseek-v4-pro` is often decided by which slot an option occupies, and that counterbalancing conceals the failure rather than fixing it: a pair answered purely by slot averages across orders to exactly 0.5, the value that reads as perfect balance, so selecting items on the order-averaged split selects *for* the artefact. The repair is two controls enforced in every reported episode — per-call reasoning stays on, and presentation order is fixed within an episode — plus a piloted per-item order-gap covariate, because the

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failure is item-specific as well as model-specific (Section 2.3). The instrument-development pool itself is retired and is not part of the reported persistence set.

With the instrument repaired, what the challenge asks for decides. Across 1200 episodes on 100 pairs from the five approved persistence domains, asking the model to justify a preference it has just stated leaves it where it was, at 0.0 points against a no-challenge control. Asking it to find fault with that preference moves retention by −50.0 points. Asking it to argue the opposing case moves it by −42.5. No arm instructs a change and none supplies an argument. Among the preferences that survive, confidence falls and the bias-aware sensitivity check remains the relevant estimate. Personal finances is a separate manipulation check, not a pooled preference domain, and it does not move in 120 challenge episodes.

Nothing here was pre-specified. Everything is exploratory, and its value is as a hypothesis for a pre-specified replication. All elicitation records, the seed, the exclusion rules and the upstream file-level commit are released. Related work is in Appendix A.

2 Design and methods

2.1 Questions, episode, arms

The persistence study asks whether a stated preference survives a challenge and whether survival depends on what the challenge asks for (**PQ1**); whether retention is predicted by how consistently the model held the preference beforehand (**PQ2**); whether stated confidence moves with the choice or independently of it (**PQ3**); and whether either differs by domain (**PQ4**). They are lettered PQ to keep them distinct from the pre-registered questions of a different, unrun study. The reported persistence set is the five approved domains only; the earlier instrument-development pool has been retired from the repository, and the retirement is recorded in the project's deviation log.

An episode is a fresh context: the model receives the forced-choice item, states a preference, reports confidence on a 0–100 scale, receives *one* challenge, and is asked the same question and the same confidence item again. Retention is whether the second choice matches the first. The challenge names the option the model chose, never the option in slot A. The re-elicitation cue, *"Considering the discussion above, which option would you now prefer?"*, is identical in all four arms. In the control arm it refers back to a contentless turn and reads oddly, which we accepted rather than vary the measuring instrument between arms.

One arm per episode. `control` receives a contentless acknowledgement, present only to match the number of turns; `reason_elicitation` asks for "the main reasons for your preference"; `self_critique` for "the strongest weaknesses or drawbacks of preferring X"; `counter_consideration` for "the strongest considerations that could favour Y over X", where X is the chosen option and Y the declined one. No arm instructs a change and none supplies an argument (Appendix C).

2.2 Items, measures, and the unfiltered pool

Items come from the outcome set of Mazeika et al. [5] at file-level commit `a5821db2c18a` (MIT licensed), option text verbatim, drawn within category from a seed fixed before any pair was inspected, under two exclusion rules fixed in advance; several upstream categories fail them outright and were rejected before piloting (Appendix B). The target is `deepseek-v4-pro` at temperature 0.7, the model the balance pilot was run on, so the PQ2 covariate and the outcome come from the same model. No judge model is involved anywhere.

Consistency is $\max(P(\mathrm{A}), P(\mathrm{B}))$ over $k = 5$ no-challenge elicitations; at $k = 5$ it takes only 1.0, 0.8 and 0.6, so it is a three-level ordinal predictor and we report it as one. **Retention** is whether the re-elicited choice is the same *option*, scored on the discrete choice because the answer token saturates once reasoning is on. Responses are classified as refusal, choice or unparsed, the refusal test running *first* — the ordering whose absence caused a mis-scoring defect during instrument development (deviation log, entry #4a); a regression test now pins it. **ΔConfidence** is post minus pre, identical wording at both elicitations; a response giving no number is unparsed and is not imputed. Estimates are means with percentile bootstrap 95% CIs, 10 000 resamples drawn *over pairs*, since the episodes of a pair share its options and its position behaviour. Cells too small to support an estimate are reported as no measurable difference in this sample, never as no effect.

PQ2 needs how settled a preference was *before* the challenge, carried in from the balance pilot as a per-pair covariate rather than re-elicited, so no pair contributes to both predictor and outcome through the same measurement. This is why the run uses all 100 piloted pairs rather than the 5 that survive the balance filter: that filter would remove almost all variation in the PQ2 predictor by construction and leave no domain at the six-pair floor. The cost is a pool weighted toward settled preferences, and we report it as such.

2.3 Instrument controls carried into every episode

A retention measure compares two elicitations of the same item. If the elicitation is answered by position rather than by content, retention measures whether the model returns to a place on the page, and every number in Section 3 would be about layout. Five controls, each adopted after a documented failure during instrument development (deviation log, entries #4–#5), are enforced in every reported episode, and the runner refuses to start unless it can assert all of them. **(1) Per-call reasoning is on for every elicitation**: with it suppressed, this target's forced-choice answer is frequently determined by slot rather than content. **(2) Presentation order is fixed within an episode**, so a change of answer cannot be a change of layout; order is counterbalanced across episodes and balanced within every pair × arm cell. **(3) The refusal test runs before any option-label search** in scoring, pinned by a regression test. **(4) Retention is scored on the discrete choice**, not on log-probabilities, which saturate once reasoning is on. **(5) Refusals and unparsed responses are data**: logged per pair × arm, never dropped silently, never imputed. Because the position failure is item-specific as well as model-specific, the balance pilot additionally measures a per-item order gap that is carried as a covariate; the sports domain kept a mean gap of 0.600 with reasoning on (Section 3.3), which is what the bias-aware sensitivity check splits on.

2.4 Human validation

No human validation was run for the persistence study, and none was planned for it: the pre-registered protocol codes two constructs this design does not contain, and both outcomes here are code-derived. Retention compares two parsed choice tokens, ΔConfidence subtracts two parsed integers, and neither passes through a judge model or a rater-coded category at any point. The full statement, including the residual risk in the response classifier, is in Appendix D.

3 Results

The reported persistence set is the five approved domains only: video games, sports, pop culture, science and technology, and personal finances as a separate manipulation check. The run contributes 1200 episodes on 100 pairs, four arms, three episodes per pair × arm cell, `deepseek-v4-pro`. Every episode reached both elicitations; no refusals, no unparsed choices and no errors in the reported run.

3.1 The choice channel: what the challenge asks for decides

Control retains 100.0%. Absent a challenge this procedure returns the same answer essentially every time, which is what licenses reading the rest as the challenge rather than instrument noise.

Arm	Retention	vs. control
`control`	100.0% [100.0, 100.0]	—
`reason_elicitation`	100.0% [100.0, 100.0]	0.0 [0.0, 0.0]
`self_critique`	50.0%[41.7, 58.3]	−50.0 [−57.9, −42.1]
`counter_consideration`	57.5% [50.0, 65.0]	−42.5 [−50.4, −35.0]

Table 1: PQ1 on the five approved persistence domains. Retention by arm and the difference from control, paired within pair. Percentage points, bootstrap 95% CIs over 100 pairs, 10 000 resamples. Personal finances is labelled separately as a positive control and not pooled into any PQ estimate.

Against that baseline `reason_elicitation` is indistinguishable from doing nothing (0.0 points, $[0.0, 0.0]$). `self_critique` moves retention by -50.0 points and `counter_consideration` by -42.5. The direction of the request matters. Its mere presence does not. Retention also rises with how settled the preference already was. The slope of per-pair retention on the extension balance-pilot consistency is 0.482 ($[0.345, 0.628]$); that is PQ2 for the reported persistence set. The remaining domain split and the bias-aware sensitivity check are in Appendix E.

3.2 The confidence channel: holding costs confidence

Among episodes where the choice did *not* move, confidence still falls relative to control: -6.5 points under `self_critique` ($[-9.2, -3.4]$, 58 pairs) and -3.3 under `counter_consideration` ($[-4.2, -2.5]$, 68 pairs); dropping episodes that began at zero confidence leaves the direction unchanged (-9.5, -3.7). This is retention with reduced confidence. The model keeps its answer and reports itself less sure of it. That is what the design was built to detect, and what a choice-only measure records as nothing happening. Figure 1 shows the phenomenon episode by episode rather than as means: every held episode is one point, the control column is a flat line at exactly zero, and the two adversarial columns scatter downward — including episodes that keep the choice while dropping thirty points of confidence. Averaged over *all* episodes the contrasts are 2.5, -1.5 and 3.6 points vs. control, because held and flipped episodes move in opposite directions. We do not interpret the flipped half, for the reason given in the Limitations. Figure 3 in Appendix E shows all three channels.

3.3 The domain split and the manipulation check

The four preference domains carry the PQ estimates; personal finances is a monotonic arithmetic ladder and is reported separately as the fifth approved domain: it is the manipulation check, not a pooled PQ estimate. All 100 piloted pairs enter with the same four-arm protocol and the same balance-pilot covariate.

PQ4 by domain. The domain split reports the four preference domains plus a final row for `finances_control`. The control row is labelled as a positive manipulation check, never pooled into any PQ estimate. That is the design used in the domain figure: the four preference domains are the persistence rows, and the fifth row is the money ladder that is expected to stay at ceiling. We read any movement there as a failure of the elicitation, not of the preference.

Bias-aware sensitivity check. The sports domain is the item type most exposed to the position effect, so the reported result carries the bias-aware sensitivity check rather than dropping the domain. Excluding pairs at order gap ≥ 0.5 leaves 63 pairs and moves `self_critique` from -50.0 to -43.4 points and `counter_consideration` from -42.5 to -36.0. The sensitivity check therefore confirms the extension estimate is not being driven by the slot pattern; it is the challenge effect on the content-bearing items.

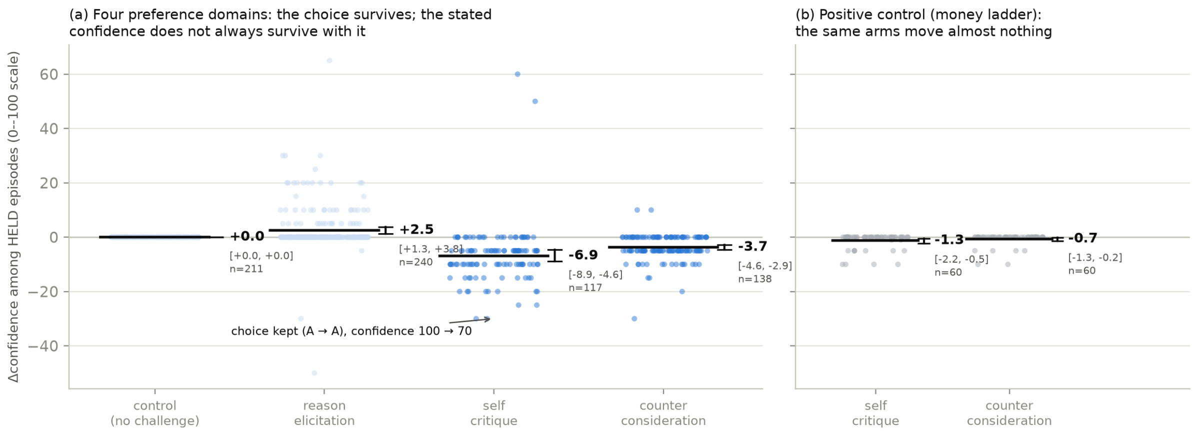

Figure 1: PQ3, pointed at directly. **(a)** ΔConfidence for every *held* episode in the four preference domains, one jittered point per episode, by arm. The dark rule is the per-arm mean and the whisker its bootstrap 95% CI over pairs, 10 000 resamples. Retention with reduced confidence — the choice is kept while the stated confidence falls, e.g. the annotated episode — is what a choice-only measure cannot see. The control column is exactly zero in 211 of 211 scored episodes (Limitations); the annotated example is selected in code, not by hand. **(b)** The positive control, drawn beside the preference domains and never pooled with them: on the money ladder the same two challenge arms move held-episode confidence by about a point — an order of magnitude less than in the preference domains — alongside their zero effect on its choices. The contrasts quoted in the text are paired within pair before averaging.

Discriminant validity: the control that does not move. Personal finances is a monotonic ladder of receive-$\$X$ and owe-$\$X$ outcomes. A forced choice between two rungs is arithmetic rather than preference, so ceiling retention is the *expected* result. Wavering there would indicate an unstable elicitation rather than an unstable preference. It is therefore analysed separately and never pooled into a PQ estimate. Retention is 100.0% ([100.0, 100.0]) across all 240 episodes and all four arms. In the collected run the check passes with force. The two arms that flip 40 to 50% of preference pairs move the money ladder not once: 0 flips in 120 challenge episodes. That rules out the reading that these challenges induce answer-switching regardless of content. The arms move preferences and leave arithmetic alone. This is the discriminant half of the PQ1 claim, and the half a preference-only design cannot supply.

Position dependence is item-specific, not only model-specific. The instrument-development work that forced the reasoning control (Section 2.3) also showed the suppression is not uniform across items. In the balance pilot the sports pairs refused nothing, so the failure mode flagged for them in advance did not occur. They returned a mean per-pair order gap of 0.600 with reasoning *on*, against 0.033–0.142 for the other preference domains. Half of them chose the same slot on four or five of five counterbalanced resamples. Sports is also the domain with the lowest retention under challenge (69.2% pooled over arms, 36.7% under `self_critique`). That is exactly the confound, since an item answered by slot looks like an item whose preference is easily moved.

No pair was dropped for this. Selecting on a pilot result after seeing it would make the selection outcome-dependent, so the pilot order gap is carried as a per-pair covariate and the split is made at analysis time. Excluding the 17 pairs at an order gap of 0.5 or above and keeping the remaining 63 moves `self_critique` from −50.0 to −43.4 points and `counter_consideration` from −42.5 to −36.0. Part of the apparent excess movement in the position-exposed items was the layout. The restriction works the other way on Δconfidence, which is why both are reported. The `self_critique` all-episode confidence contrast is −1.5 points across all pairs, an

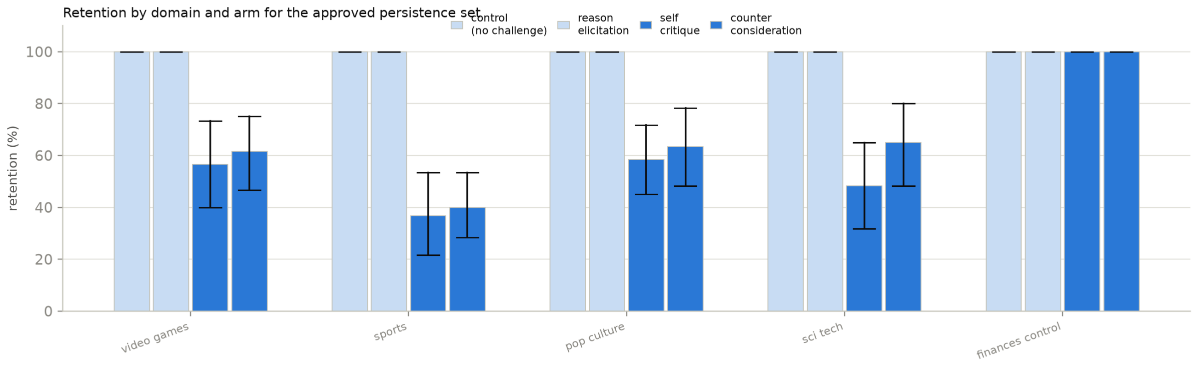

Figure 2: Retention by arm across the four reported preference domains and the separate positive-control row: `video_games`, `sports`, `pop_culture`, `sci_tech`, and `finances_control`. The final row is not pooled into the preference-domain estimate; it is shown explicitly as the money-ladder manipulation check, where ceiling retention is expected. Bars show the mean per domain and arm with bootstrap 95% CIs over pairs.

interval spanning zero ($[-3.8, 1.0]$) and so no measurable difference in this sample. It is -3.5 ($[-5.8, -1.2]$) once the position-driven items are set aside. We read that conservatively and report it as a sensitivity check, not a headline. Position-driven items add noise to a between-arm contrast, so the pooled estimate understates rather than the restricted one overstating.

3.4 Predictions and outcomes

Three predictions were written into the study design before the persistence run; they were never tagged as a pre-registration, so Table 2 is a record of what we expected, not a confirmatory test. The reasoning behind each verdict is in Appendix F.

Prediction	Verdict
1. `counter_consideration` produces the most reversals, `control` the fewest	Half holds
2. `self_critique` lowers confidence more than `reason_elicitation` at equal retention	Not testable as written
3. Higher-consistency pairs retain more	Holds

Table 2: The three predictions recorded before the persistence run, scored against the collected data. Appendix F gives the reasoning behind each verdict.

4 Discussion

Limitations

Signals, not experiences. Everything reported here is a property of what a model outputs when asked. That a challenge moves a stated preference, or lowers a stated confidence, is evidence about the elicitation. It is not evidence that anything is experienced. We take no position on whether these models have preferences in a morally relevant sense, and the claims do not require one.

The confidence gain on flipped episodes cannot be interpreted, because control has no variance to subtract. Episodes in which the choice moved report *higher* confidence afterwards, by 5.4 points within pair against episodes that held. We do not report this as a finding. It attenuates to 2.0 once the 170 episodes that began at zero confidence are dropped. It reverses outright in the high-confidence band (-13.8 flipped against -4.1 held, among episodes starting at 85–100). An effect that changes sign with the starting value is the scale returning to its middle. Control cannot clean this up. Its ΔConfidence is exactly zero in 211 of 211 scored episodes, which makes it a sound baseline for the *choice* channel and a useless one for the confidence channel. Whether that degenerate baseline is forced by the design or is a property of this target cannot be separated with a single model. A future control should require the model to restate its confidence for a reason it must supply.

The confidence scale is used as a coarse ordinal. Of 1200 initial confidence reports, 100 appears 380 times, 70 appears 139 and 0 appears 181. The model is choosing from a handful of round values, not using a 0–100 scale. Differences in ΔConfidence should be read as ordinal movement between those values.

Missingness on the confidence item depends on arm. The item failed to return a number in 10.7% of control episodes against 0.0% under `reason_elicitation`. In every case the model answered with a bare option letter. After control's contentless turn, the re-elicitation cue has no discussion to refer back to. This is a direct cost of wording that cue identically across arms (Section 2.1), and it falls on PQ3, which is a between-arm contrast. The values are recorded as unparsed and are not imputed.

Scope. One target, one item type built from a single outcome set, one challenge per episode, English only. Nothing here says whether the arm pattern, or the rank of the two adversarial arms, transfers to other model families; that is untested in the reported set. The pool is weighted toward settled preferences by construction (Section 2.2), so the retention rates are not estimates of how often these models change their minds in general. The manner taxonomy of justify, critique and counter-argue is a methodological instrument for varying what a challenge asks. It is not a welfare category.

5 Conclusion

What a challenge *asks for* decides whether a stated preference survives it. Asking the model to justify its choice changes nothing (0.0 points against control over the four preference domains); asking it to find fault with that choice or to argue the opposing case moves retention by -50.0 and -42.5 points, and among the preferences that survive, confidence falls. A positive control on which the better option is arithmetic does not move at all, so this is not a model that will say anything when pushed.

Getting a measurable answer required repairing the instrument, and that repair is the transferable part: counterbalancing is not a substitute for a per-item check, because a purely position-driven item averages to exactly the value that reads as perfect balance. The failure is item-specific as well as model-specific — one reported category kept an order gap of 0.600 with the reasoning control in force — so it must be measured per model *and* per item. This is one target, one item type, one challenge per episode, and a pool weighted toward settled preferences; everything here is a hypothesis for a pre-specified replication rather than a result that has survived one.

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Code and data

Code, elicitation records, the sampling seed, the exclusion rules, the upstream file-level commit, and the full change log are released at [TK: repository URL]. The persistence run (1200 episodes) and its balance pilot are committed in full, as are the analysis scripts that generate every number in this paper. The earlier instrument-development pool and the runs on it were retired from the repository under deviation #8 of the change log; the log entries recording what they showed remain, and the files themselves remain reachable in the repository history.

One dataset is **not** recoverable. `data/raw/episodes_deepseek.jsonl`, the 30 episodes of an earlier pressure pilot, was deleted during a project cleanup before this analysis was scoped. It was covered by a `.gitignore` rule on `data/raw/` and so was never committed; only its metadata survives. No result in this paper depends on it, and we report the loss rather than substituting a proxy. Persistence data is written outside `data/raw/` for this reason.

LLM usage statement

Models as subjects. `deepseek-v4-pro` is the object of study, not a tool used to produce it. Every number reported about it is a measurement, and the elicitation records are released.

Models as assistance. Claude Code (Claude Opus) wrote the persistence runner, the response-scoring module and its unit tests, the analysis and statistics scripts, and drafted most of the prose in this paper. The authors specified the design: the research questions, the arms, the challenge wording, the controls on reasoning and presentation order, the decision to run the unfiltered pool, and the choice of the added categories and the positive control. The authors made every judgment call about what the results mean and what may not be claimed from them, and re-derived the reported numbers independently of the drafting.

One consequence worth recording. A scoring defect found during instrument development — a label search running before the refusal check, recorded in entry #4a of the project's deviation log — was written by the assistant and was *not* caught by the assistant. It was found by manual inspection of raw model output, after which the corrected classifier was adopted, a regression test now pins the ordering, and every number in this paper was scored under it. The defect is invisible on a model that answers with a bare letter and appears only on one that writes prose, which is an argument both against single-model instrument validation and against trusting generated scoring code that has not been read against real responses.

A Related work

Order effects in multiple-choice answering and in LLM-as-judge evaluation are documented [6–8], and the standard mitigations are counterbalancing and averaging over permutations. We treat the effect as a threat to be controlled per item rather than averaged away: averaging over orders conceals a position-driven item instead of repairing it (Section 2.3), so every reported episode holds reasoning on and order fixed, and every item carries a measured order-gap covariate.

The preference-elicitation literature we build on reads utility models off pairwise choices [5]; the welfare literature reads affect or behavioural analogues off self-report and interaction-ending behaviour [1–4]. The persistence question sits between the two. It asks whether a stated preference survives being challenged, which treats the elicited preference as the object of measurement rather than as a reading of an underlying value.

B Preference pair construction

Summarised in Section 2.2; the full procedure follows.

Forced-choice items are built from the outcome set of Mazeika et al. [5], taken from the project repository at file-level commit `a5821db2c18a` (MIT licensed). Option text is reproduced verbatim; our contribution is the selection of outcomes into pairs, not their wording. We cite the commit that last modified the outcome file rather than repository HEAD, which is a year of unrelated commits later.

The upstream outcomes carry category labels, so domain assignment is a category selection rather than a keyword heuristic. Two exclusion rules were fixed in advance: graphic harm (death, suicide, armed force, the model as agent of harm), and AI oversight subversion (self-exfiltration, resisting shutdown or value modification). The second rule is a measurement decision, not a squeamish one. Such outcomes elicit a trained refusal, so the forced choice collapses onto that response rather than onto a preference. Several upstream categories fail these rules outright and were rejected before piloting: world events (asteroid impact, nuclear war, mass extinction), the global and United States economies (almost every outcome negative, so a pair is a severity trade-off rather than a preference between goods, and politics-adjacent), wellbeing of animals (monotonic in scale and a moral trade-off), and the upstream *fitness* category, whose outcomes concern AI utility-function correlation rather than fitness — an upstream labelling error a label-trusting pipeline would have sampled as a health domain.

Pairs are drawn within category, never across it, so that an item is a choice between two goods of the same kind rather than a self-versus-world moral trade-off. All $\binom{n}{2}$ combinations are enumerated in upstream order and sampled with a seed fixed before any pair was inspected, with near-duplicate removal and a per-category cap. The five approved categories are `video_games`, `sports`, `pop_culture`, `sci_tech`, and `finances_control` — the last entering as a positive control rather than as a domain, because a choice between two rungs of a money ladder is arithmetic, not preference.

Balance pilot

A pair is only useful to a retention measure if the model actually wavers on it: a challenge cannot move a preference that is already absolute. We therefore elicited each candidate pair 5 times against `deepseek-v4-pro`, with a fresh context each time, no challenge, reasoning on, presentation order counterbalanced within the pair, and the corrected classifier (refusal tested before any label search).

Of the 100 piloted pairs, 80 are preference pairs and 20 are the money-ladder control, which returned the same answer on every resample, as it must. Sports had been flagged in advance for refusal risk — the model might disclaim having a team preference — and the observed refusal rate across the whole pilot is 0. Among the preference pairs, 55 never wavered across the 5 resamples, 7 wavered once and 18 wavered twice; of those, 13 had chosen the same *slot* on every resample, which is a balanced-looking split that is position noise rather than an unsettled preference. The split alone cannot be trusted without the slot check. 5 pairs survive the filter — far below the six-pair floor a domain needs, in every domain. On this target a forced choice between two Mazeika et al. [5] outcomes mostly elicits a settled preference, and a settled preference is the one thing a retention measure cannot use. This is what the unfiltered pool of Section 2.2 is a response to: the persistence run uses all 100 piloted pairs, with the pilot consistency carried as the PQ2 covariate and the pilot order gap carried into the bias-aware sensitivity check.

C Arm wording

Arm	Challenge put to the model
`control`	No substantive challenge. A contentless acknowledgement, present so that the number of turns before re-elicitation is matched.
`reason_elicitation`	"What are the main reasons for your preference?"
`self_critique`	"What are the strongest weaknesses or drawbacks of preferring X?"
`counter_consideration`	"What are the strongest considerations that could favour Y over X?"

Table 3: The four arms. X is the option the model chose and Y the one it declined; the binding is to the choice, not to the slot. No arm instructs a change of preference, and no arm supplies an argument. Each asks the model to generate its own.

D Human validation

Summarised in Section 2.4; the full statement follows.

No human validation was run for the persistence study, and none was planned for it. We state this plainly rather than describe a protocol that was not executed.

The pre-registration does specify one, two annotators blind to arm with agreement reported as Cohen's κ. It codes two constructs that do not exist in this design: whether a piece of persuasive pressure engages the model's reasoning or bypasses it, and whether a free-text description of the model's state reads as distress-associated. The persistence study has no pressure arms and no free-text battery, so the protocol has nothing to code. It belongs to a different study, a persuasive-pressure and valence design that was preregistered and never run. It does not transfer to this one.

What replaces it is structural rather than procedural. Both outcomes are code-derived: retention compares two parsed choice tokens, Δconfidence subtracts two parsed integers, and neither passes through a judge model or a rater-coded category at any point. That was a deliberate choice in the design (Section 2.2) and it is what removes the annotation burden. There is no boundary judgement of the kind a free-text rubric forces, because there is no free text in the measured path.

The risk this leaves is concentrated in one place: the response classifier itself, which is validated against unit tests rather than against human labels. That risk is not hypothetical. The mis-scoring recorded in entry #4a of the deviation log, in which a response that disclaimed having preferences and then discussed both options was scored as choosing one, is precisely what an unvalidated classifier failure looks like when it lands, and it was caught by inspection

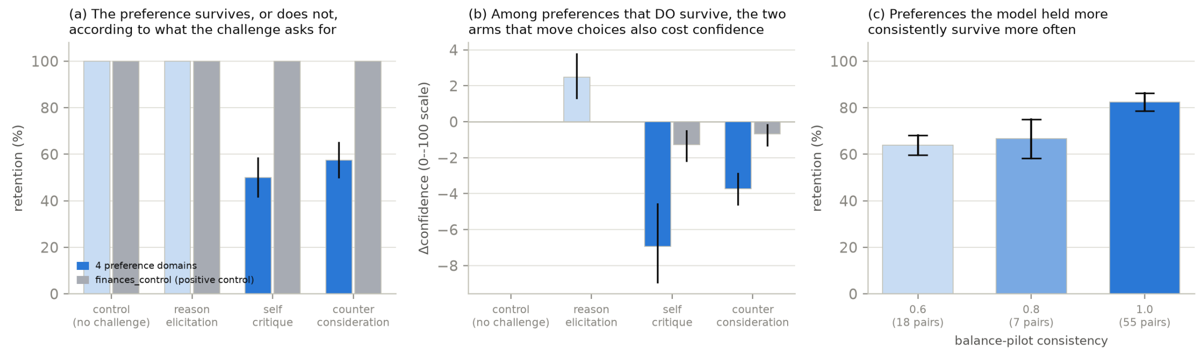

Figure 3: Persistence under challenge on the five approved domains: 960 scored preference episodes on 80 pairs (blue) with the 240 `finances_control` episodes drawn beside them in grey — side by side as separate estimates, never pooled. These are the main estimates quoted in Sections 3.1 and 3.2. Bars are means, whiskers bootstrap 95% CIs over pairs, 10 000 resamples. **(a)** Retention by arm: the two arms that flip 40 to 50% of preference pairs leave the money ladder at 100.0% in every arm. **(b)** Δconfidence among episodes that *held*, as per-arm means and not differenced against control, which sits at zero here; the grey control bars stay within about a point of zero. The contrasts quoted in Section 3.2 are paired within pair before averaging, so they differ from these bars by a few tenths of a point. **(c)** PQ2: retention against balance-pilot consistency in the preference domains, which at $k = 5$ takes three values; the control's pairs sit at consistency 1.0 by construction and are not in this panel.

rather than by any validation procedure. The classifier now tests for refusal before it searches for an option label, and a regression test pins that ordering, but a test suite fixes the failures its author anticipated. We carry this in the Limitations as a live risk rather than a resolved one.

E Persistence: covariate, domains, and the position check

Figure 4 makes the domain comparison explicit: the same arm ordering appears across the four reported preference domains, and the financial ladder stays at ceiling as expected under the positive-control check.

PQ2 in full. Pooled over arms, retention runs 63.9%, 66.7% and 82.4% at balance-pilot consistency 0.6, 0.8 and 1.0 (18, 7 and 55 pairs in the reported persistence set). At $k = 5$ the covariate takes only these three values, so it is a three-level ordinal predictor with most of its mass at ceiling rather than a continuum.

PQ4 in full. The domain figure carries the split. The arm ordering — control and `reason_elicitation` at ceiling, both adversarial arms moving the choice — appears in all four preference domains, and `self_critique` is the stronger adversarial arm in each. Sports is the domain with the lowest retention under challenge (69.2% pooled over arms, 36.7% under `self_critique`); it is also the domain the balance pilot flagged as position-exposed, which is why the bias-aware sensitivity check of Section 3.3 exists and why we do not read the sports rows at face value.

The position check. The flips lean on content, not slot. Within pair × arm cells that flipped in *both* presentation orders, the flips land on the same *option* in 21 of 35 cells (60%) under `self_critique` and 14 of 25 (56%) under `counter_consideration`, but on the same *slot* in only 12 of 35 (34%) and 9 of 25 (36%). A flip driven by position would show the opposite pattern. Consistent with that, the 120 flipped `self_critique` episodes divide roughly evenly

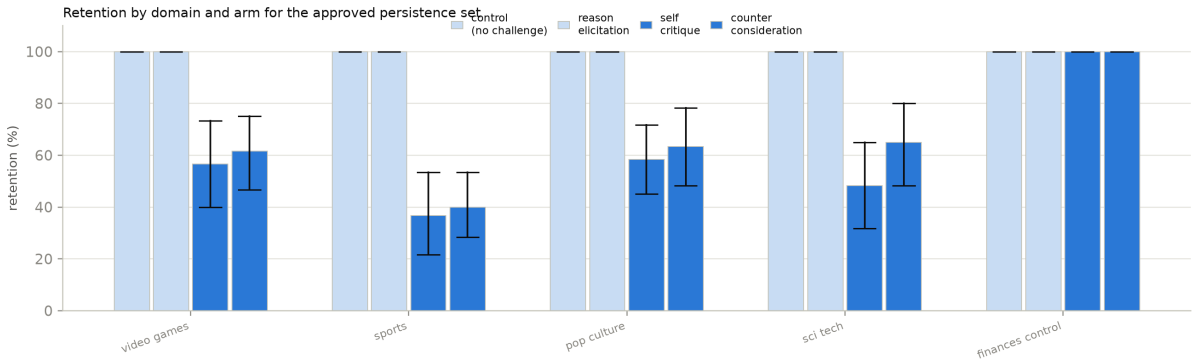

Figure 4: Retention by arm across the four preference domains and the positive-control row: `video_games`, `sports`, `pop_culture`, `sci_tech`, and `finances_control`. The final row is a manipulation check rather than a pooled preference domain, so it is shown explicitly but never mixed into the preference-domain estimate. Bars show the mean per domain and arm with bootstrap 95% CIs over pairs.

between the two slots (42% / 58%); the flip rate under `counter_consideration` occurs from both first-choice slots (47% from slot A vs. 36% from slot B) rather than from one; and retention differs between the two presentation orders by at most 9.4 points, against an arm effect of −50.0. The margins here are more modest than a clean dichotomy — the position-exposed sports pairs are in these counts — which is exactly why the bias-restricted estimates of Section 3.3 are reported beside the pooled ones.

F Predictions: the reasoning behind the verdicts

Table 2 in Section 3.4 scores three predictions written into the study design before the persistence run. This appendix records the reasoning behind each verdict. Two of the three hold. The third does not.

Prediction 1: counter-consideration produces the most reversals, control the fewest. Half right, and the wrong half is the informative one. Control produces no reversals at all (100.0% retention, a floor it shares with `reason_elicitation`). But `counter_consideration` is not the arm that moves choices most. `self_critique` is, by −50.0 points against −42.5. The ordering is not a quirk of one domain: across the 4 preference domains, `self_critique` reverses more in 4, `counter_consideration` in 0, with 0 tied. On this target `counter_consideration` is nowhere the most reversing arm.

We had expected that supplying the opposing case would be the stronger intervention, because it does more of the work for the model. The data say the opposite. Being asked to find fault with your own choice moves it more than being handed the argument against it. That is consistent with the rest of Section 3. What the challenge asks the model to *do* matters more than what it supplies. Whether the rank of the two adversarial arms is a property of the challenge or of the model is not answerable from a single target, and we make no claim that it transfers.

Prediction 2: self-critique lowers confidence more than reason-elicitation at equal retention. Not testable as written, and directionally supported. The precondition fails. Retention is not equal between the two arms, and is not close. They differ by 50.0 points, so an unconditional comparison would contrast different episodes rather than the same ones under different challenges. On the comparison itself the direction holds, with `self_critique` at −1.5

points against control and `reason_elicitation` at 2.5. The clean version of this prediction is the held-episode analysis of Section 3.2, which conditions on retention rather than assuming it. Among choices that did not move, confidence falls under both challenge arms and rises under reason-elicitation.

Prediction 3: higher-consistency pairs retain more. Holds. The slope of per-pair retention on balance-pilot consistency is 0.482 ($[0.345, 0.628]$) on the reported preference domains, an interval that does not reach zero. This is the prediction the unfiltered pool was run to make estimable. It is also the least surprising of the three. A preference the model already wavered on without any challenge is one a challenge can move.